

Reducing Tailings Risk and Unlocking Value through Scalable 3D Subsurface Sensing.

VISTA — Volumetric Integrated Sensing for Tailings Assessment

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A scalable, multi-physics sensing architecture that enables spatially continuous 3D subsurface understanding of tailings systems by integrating radar, spectral sensing, and site-specific calibration modalities.

Across the global mining industry, tailings storage facilities represent both one of the highest-consequence **safety¹ risks** and a largely untapped source of **critical minerals (CM)**. These systems are increasing in scale as ore grades decline, yet their internal structure—water distribution, layering, and material composition—remains poorly understood. This challenge sits at the intersection of critical mineral supply, infrastructure risk, and national security.

The urgency for new CM resources is increasing as the U.S. remains 100% import reliant for 12 critical minerals, while global demand is projected to triple by 2030. The scalable sensing approaches described here, based on technologies adapted from off-world exploration, offer a rapid pathway to **accelerate CM exploration** and subsurface characterization **by orders of magnitude**.

We cannot see inside tailings systems at scale. Current monitoring approaches rely on sparse, point-based measurements—such as drilling, piezometers, surface monitoring systems, and instrumentation concentrated around embankments and critical infrastructure—to infer conditions across large and heterogeneous deposits. These methods are slow, expensive, and fundamentally incapable of resolving continuous three-dimensional structure, leaving key features such as saturation zones, weak layers, and material variability undetected until deformation or failure occurs.

This limitation creates both systemic risk and missed economic opportunity. Tailings failures result in multi-billion-dollar losses and significant environmental and human impact, while tailings deposits represent viable secondary sources of critical minerals. Without reliable subsurface characterization, operators cannot confidently assess stability or determine which deposits justify reprocessing, constraining both safety outcomes and value recovery at scale. This program addresses this gap by enabling spatially continuous, volumetric (3D) subsurface characterization at orders-of-magnitude lower cost. At operational scale, equivalent characterization using conventional methods can cost hundreds of millions to billions of dollars and take years to decades, whereas emerging sensing approaches can deliver comparable spatial insight at >1000× lower cost and within weeks.

The core challenge is not the absence of sensing technologies, but the absence of a scalable system to integrate them into a reliable view of tailings systems. Existing approaches remain fragmented across high-cost instrumentation, lower-cost but uncalibrated sensors, and siloed research efforts, leaving no scalable, non-intrusive capability to generate continuous system-level understanding.

Several trends make this an unusually timely moment to address this challenge. Globally, there are ~8,500 tailings storage facilities (TSFs) containing ~282 Bt of material, with billions of tonnes added annually. Critical mineral supply pressures are increasing, and tailings are being reconsidered as potential secondary resources. Attention to tailings safety has intensified following repeated failures in Brazil, Africa, and elsewhere, while international standards such as the Global Industry Standard on Tailings Management have raised expectations for monitoring and accountability. In parallel, advances in radar and electromagnetic imaging, sensor miniaturization, edge computing, and machine learning have converged, creating a timely opportunity to rethink how sensing technologies are integrated and deployed in operational environments.

Finally, there is a structural gap in the innovation ecosystem. Despite \$2–4 billion per year already being spent globally on tailings monitoring, continuous system-level understanding remains absent. Existing approaches remain fragmented across academic research, industry practice, and high-cost instrumentation, limiting their ability to deliver scalable, integrated solutions. This program operates at the intersection of these gaps.

If successful, it will enable early identification of instability risks and **targeted intervention to prevent catastrophic failures**, while also supporting screening of tailings as secondary critical mineral resources and unlocking significant value (e.g., Bleida: ~\$25M sampled, ~\$120M+ total) at minimal survey cost (~0.05–0.15% of asset value). The integrated MMS + mUWB-GPR system delivers >1000× reduction in characterization cost, 50–80× faster surveys, and continuous volumetric coverage, enabling large sites (~18.8 km²) to be surveyed in ~1–2.3 months at ~\$138k compared to years and millions–billions using drilling-based approaches.

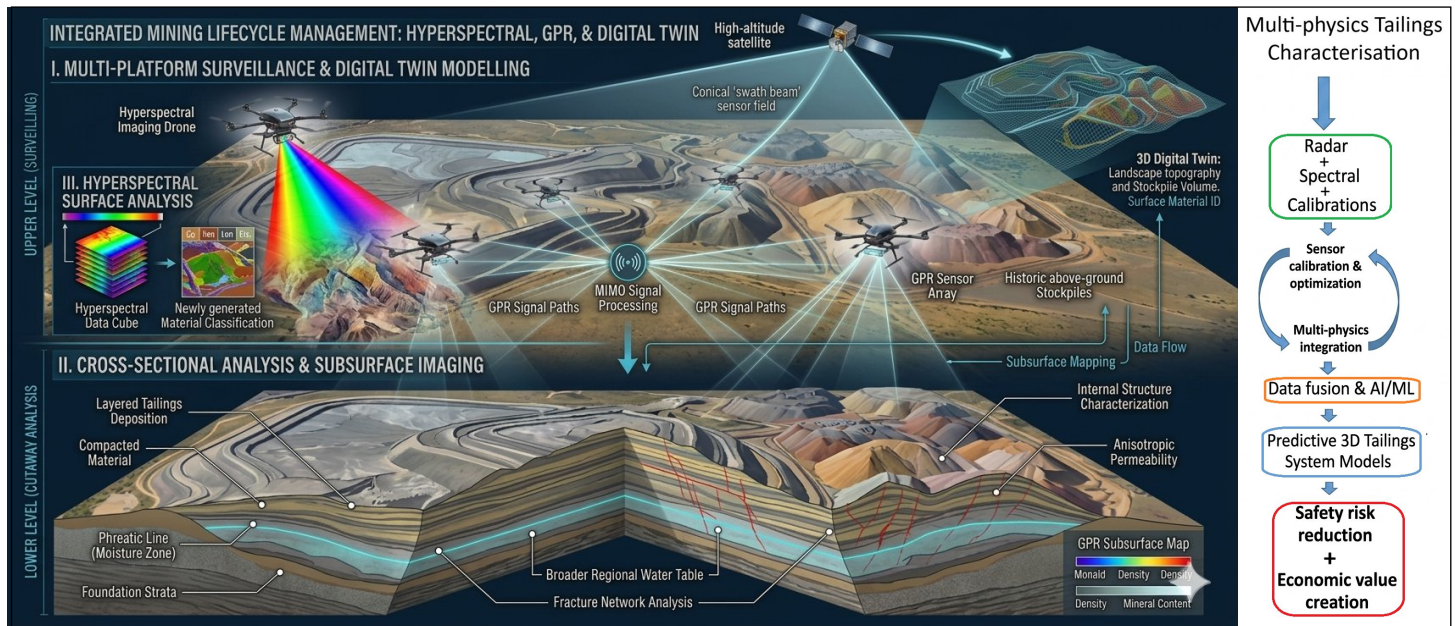


Figure 1. Multi-physics tailings characterization concept integrating spectral (MMS), radar (mUWB-GPR), and adaptively calibrated sensing.

Mining system optimization and operational value would be driven by improved material-domain understanding, enabling better routing, blending, and processing, reducing penalties and operating costs, with estimated 5-year savings of ~\$40M–\$120M (copper) and ~\$80M–\$800M (iron ore), and reaching billions (\$1.6B–\$4.0B) where mercury impurities are present. This is further enabled by **automation and scalable sensing across the mining cycle**, where MMS reduces spectral characterization costs by ~70–90% and enables distributed sensing, allowing continuous updates to resource models and improved decision-making. Beyond mining, the sensing architecture extends to defense (subsurface detection), critical minerals (proxy-based geodomaining), and emerging energy systems such as geologic hydrogen, providing dual-use capability aligned with strategic priorities.

The VISTA Approach: A Scalable Multi-Physics Sensing Architecture for Tailings Systems

This program establishes a scalable architecture for subsurface understanding by integrating advances in complementary sensing modalities—spectral (MMS / MahinaCAM) and multistatic radar (mUWB-GPR)—supported by adaptive calibrationⁱⁱ using targeted geophysical and geochemical methods (e.g., IPⁱⁱⁱ, ERT/EM, XRF). Iterative data integration enables volumetric (3D) characterization of tailings systems, shifting subsurface understanding from sparse, point-based sampling to spatially resolved system-level models for safety monitoring and CM targeting.

Two capabilities underpin the approach. First, extending mUWB-GPR performance beyond conventional penetration limits enables volumetric imaging of tailings systems at depths previously inaccessible. Second, the program proposes a new modular multispectral micro-sensor system (MMS) architecture, enabling scalable, low-cost proxy-based mineral characterization across mining environments. The program focuses on advancing these scalable sensing modalities while leveraging existing geophysical methods as calibration and validation constraints (LTM 1).

Modular Micro-Sensor Spectral System (MMS) is a compact, configurable multispectral sensing approach derived from hyperspectral data but restricted to diagnostic wavelengths. Hyperspectral sensing enables identification of diagnostic mineralogical and alteration-related spectral features across surfaces and drill core. Calibrated against high-resolution hyperspectral reference datasets, MMS preserves material-domain and mineralogical characterization capability while reducing system complexity and cost (up to ~10×), enabling scalable, distributed deployment. It provides high-resolution mineral identification on surfaces and cores and supports calibration of subsurface sensing and proxy-based CM screening (LTM 1).

Multistatic Ultra-Wideband Ground-Penetrating Radar (mUWB-GPR) enables volumetric imaging of subsurface structure by transmitting and receiving radar signals from multiple positions to reconstruct three-dimensional domains. The program extends radar performance beyond conventional limits through multistatic configurations, antenna design, and multi-frequency operation, targeting penetration from typical ~3–10 m to ≥70 m in dry legacy tailings (and ~30 m in ~20% VWC conditions). This enables mapping of internal layering, density contrasts, voids, and volumetric water content, providing continuous structural models of tailings systems (LTM 2).

Success will be demonstrated through staged quantitative performance metrics including: (a) 3–4 targeted MMS micro-sensors reproduce hyperspectral mineral detection performance for target minerals; (b) mUWB-GPR delivers reliable structural imaging to ≥30 m in ~20% VWC materials and ≥70 m in dry legacy tailings; (c) integrated MMS + mUWB-GPR systems resolve material domains and mineralogical proxies relevant to CM recovery to ≥20 m depth (mid-program ~10 m); (d) detection of subsurface structural domains associated with liquefaction potential, saturation zones, and weak layers to ≥75 m (mid-program ~35 m); (e) >10× reduction in spectral sensing cost, ≥10× increase in spatial coverage relative to point-based approaches, and survey of large sites within days–weeks; and (f) integration of MMS, mUWB-GPR, and calibration datasets into predictive 3D models validated against ground-truth test sites.

Program cost: \$27–\$44MM over five years. The program focuses on integrating and validating sensing architectures in real-world environments and is well suited to a hybrid funding model spanning government agencies, industry partners, and philanthropic capital (Appendices 2-4). A multi-partner deployment model aligned with industry practice will be established^{iv}.

The majority of funding will support field validation, system integration, and operational deployment rather than core sensor development. The program supports a focused portfolio of coordinated research and field-validation projects, including:

- Tailings Test-Site Partnerships^v and Ground-Truth Dataset Development (*2-3 sites for benchmarking*)
- mUWB-GPR System Optimization for Tailings Environments (*extend depth and imaging capabilities*)
- Hyperspectral Calibration for MMS Mineral Domains and CM Indicators (*for micro-sensor selection*)
- Induced polarization (IP) as a Calibration and Validation Layer (*targeted IP for interpretation*)
- Integrated Multi-Physics Sensing Prototypes (*MMS + mUWB-GPR adaptively calibrated field systems^{vi}*)
- Data Fusion and Predictive 3D Tailings Models (*for safety monitoring and resource screening*)
- Technology Extrapolation to Other Subsurface Resource Domains (*CM exploration, geologic H₂ / NH₃*)
- Operational Demonstration and Adoption Pathway (*staged pilot deployments*)

Expressions of interest include leading research institutions, industry partners, and subject matter experts from NASA Ames, HZDR, USC, USGS, JHU-APL, IMDEX, Channel Systems, TerraGlo, Addis Energy, Endolith, and additional groups, based on initial discussions and technical engagement. Named organizations reflect technical discussions, advisory input, or expressions of interest only and do not imply institutional endorsement, formal partnership, or funding commitment unless separately agreed.

The program is structured in phased workstreams spanning multiple stages (Figure A3.1). While presented sequentially, integration and validation activities drive continuous refinement of sensing systems, calibration approaches, and data-fusion methods. Site access, test-bed selection, and performer partnerships are established during Phase 0 through engagement with operators and research partners.

(Here, MMS refers broadly to modular multispectral sensing approaches, including MMS and MahinaCAM configurations.)

Phase	Years	R&D Milestones	Test-bed Enablers
0 — Design	0–0.5	• Calibration framework• Architecture & sensing targets• MMS spectral design• Test site selection	• 2–3 tailings sites• Core & geotechnical data• Lab calibration + hyperspectral data
1 — Proof	0.5–1	• MMS vs hyperspectral validation• mUWB-GPR imaging + VWC• Initial multi-sensor datasets	• Lab/testbeds• Calibrated materials• Benchmark datasets
2 — Integration	1–2	• MMS + mUWB-GPR workflows• Calibration modalities deployed• Integrated sensing prototypes (field-tested)• Data fusion & inversion• Predictive 3D models	• Instrumented sites• Drone/ground platforms• Operational datasets
3 — Validation	2–5	• Performance across regimes• Structural risk detection• Operational workflows	• Active + legacy sites• Operators + regulators• Repeat monitoring

Endnotes

- ⁱ In this context, “safety” refers primarily to geotechnical and environmental risk associated with instability, saturation, liquefaction potential, and structural change within tailings systems.
- ⁱⁱ Adaptive calibration refers to the selective use of complementary methods (e.g., IP, ERT/EM, seismic) as needed for a given site to constrain and refine interpretation of multi-sensor data.
- ⁱⁱⁱ Induced polarization (IP) is a geophysical method that measures the ability of subsurface materials to store and release electrical charge, providing sensitivity to mineralogy (e.g., sulfides, clays) and supporting interpretation of material composition and distribution.
- ^{iv} In mining, consortium-based models are often not effective at accelerating commercial deployment. Constraints around data sharing, intellectual property, operational risk, and site access can slow decision-making and adoption. As a result, direct, site-specific partnerships with operators are more effective for calibration, validation, and real-world deployment of integrated sensing systems.
- ^v Through Helmholtz-Zentrum Dresden-Rossendorf (HZDR), we are connected to a global mining R&D network including Nordic Strategy Partners AB | GFZ | CTTG | Babeş-Bolyai University | TheiaX GmbH | GeoKinesia | Boliden Mineral AB | Advanced Mining Technology Center | Codelco | Cupru Min S.A. | FQM Trident Ltd.

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Appendix (Supporting Materials)

- Appendix 1 – Program Cost & Economic Model
- Appendix 2 – Multi-Physics System Architecture
- Appendix 3 – Program Phasing & Deployment Roadmap
- Appendix 4 – Stakeholder Landscape & Partnerships
- Appendix 5 – Technical Notes & Sensing Modalities
- Appendix 6 – Letters of Intent

Linked Technical Materials (LTM)

- [L1 – MMS & mUWB-GPR](#)
(Advanced Geosensing Technologies for Terrestrial Mining Applications)
- [L2 – mUWB-GPR technical background](#)
(Summary Sheet: GPR tomography potential terrestrial mining applications)
- [L3 – MMS for cost savings \(penalty reduction\)](#)
(Exploration & downstream processing)
- [L4 – MMS for reaction domain mapping](#)
(Geologic H₂ & NH₃ applications)
- [L5 – Extended Program Concept](#)
(Full technical description and system architecture)

- Appendix 1 – Program Cost and Economic Model

Table A1.1 Preliminary Program Cost Envelope

Program Component	Estimated Cost
Phase 0 – Technical De-Risking & Program Design (6–12 months)	\$0.5 – \$1.5MM
A) Program Development & Validation (5-year program)	\$27– \$44MM
Technology development	10–20%
Data science & modelling	15–20%
Field pilots & test sites	25–35%
Systems integration	10–15%
Industry validation	10–15%
Program coordination	5–10%

Table A1.2 Preliminary program cost envelope showing indicative budget ranges for major components; to be refined during Phase 0 scoping and design.**Program Logic —**

Distributed, application-specific sensing → decision-grade resource confidence → critical minerals & geologic energy → Earth ↔ ISRU transition.

Preliminary Program Cost Envelope (Performer-Based, Planning Range)		
	min (\$MM)	max (\$MM)
Phase 0 – Technical De-Risking & Program Design (6–12 months)	0.5	1.5
A) Program 1 (BiTs Renaissance / “Resource Program”)	\$27- 44MM	
Core Performers (Program Execution)		
NASA Ames (RSAA) – Space transition & mission validation partner (Phase-dependent, non-funded)		
USGS – Data alignment & terrestrial validation collaboration	1	3
JHU-APL	6	8
National Labs (e.g., LBNL) – via DARPA / DOE alignment	8	12
Technical Development Performer (Lab / SME / University)	5	9
Deployment Performer (Mining Operator / Tailings Entity)	5	8
Geologic H ₂ / NH ₃ Domain (TBC)	2	4
Program 1 Core Subtotal :	27	44
Program End State —		
Validated, field-deployable resource-confidence system enabling largely drill-free characterization workflows through calibrated geophysical, spectral, and proxy-based material-domain sensing for tailings safety, critical mineral recovery screening through indirect mineralogical indicators and pathfinder systems, geologic energy systems, and off-world resource assessment.		
Strategic Co-Funding & In-Kind Expansion Layer (Non-Core):		
Mining companies: (<i>indicative: \$5–10MM in-kind</i>)		
Bankers / investors: ?		
Preliminary Program Cost Envelope – Phase 0 (v0.9 May 2026)		

• Appendix 2 – Multi-Physics System Architecture

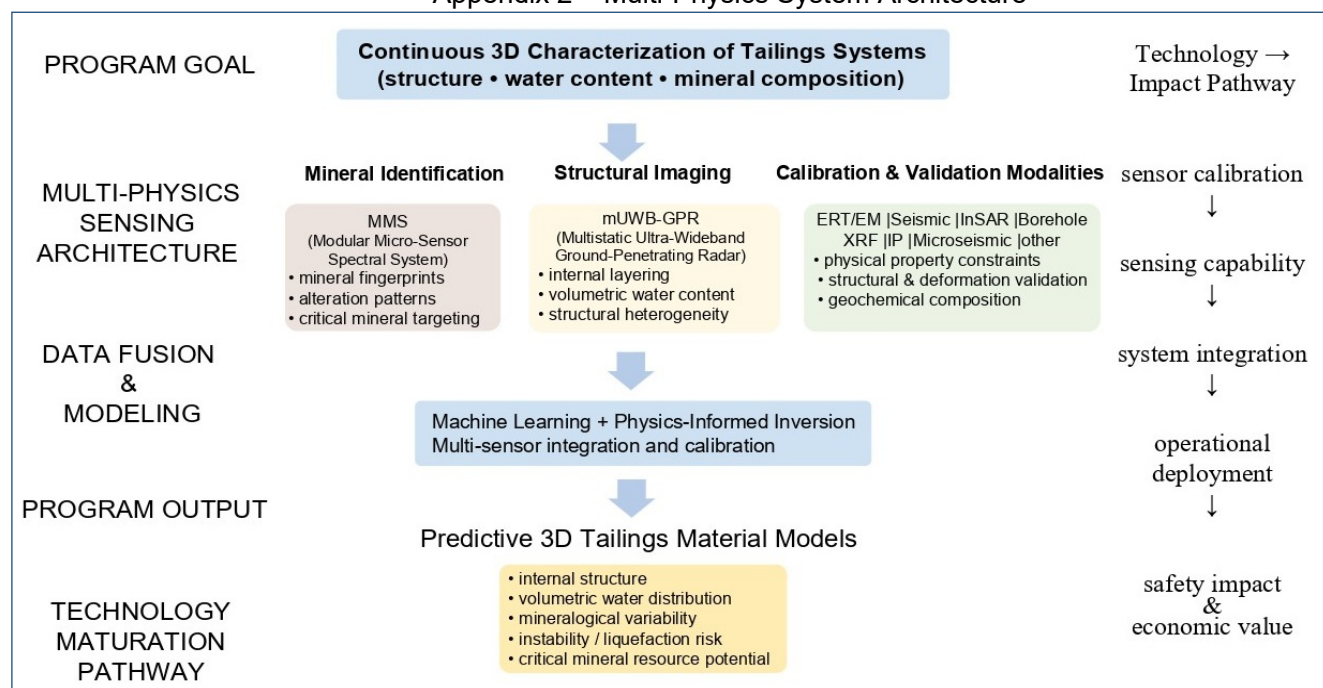


Figure A2.1 Multi-Physics Program Architecture for Volumetric Tailings Characterization. “Critical mineral detection” refers primarily to detection of mineralogical proxies and related spectral indicators associated with critical mineral systems rather than direct ore mineral detection. “Continuous” refers to spatially continuous volumetric characterization rather than persistent temporal monitoring.

• Appendix 3 – Program Phasing and Deployment Roadmap

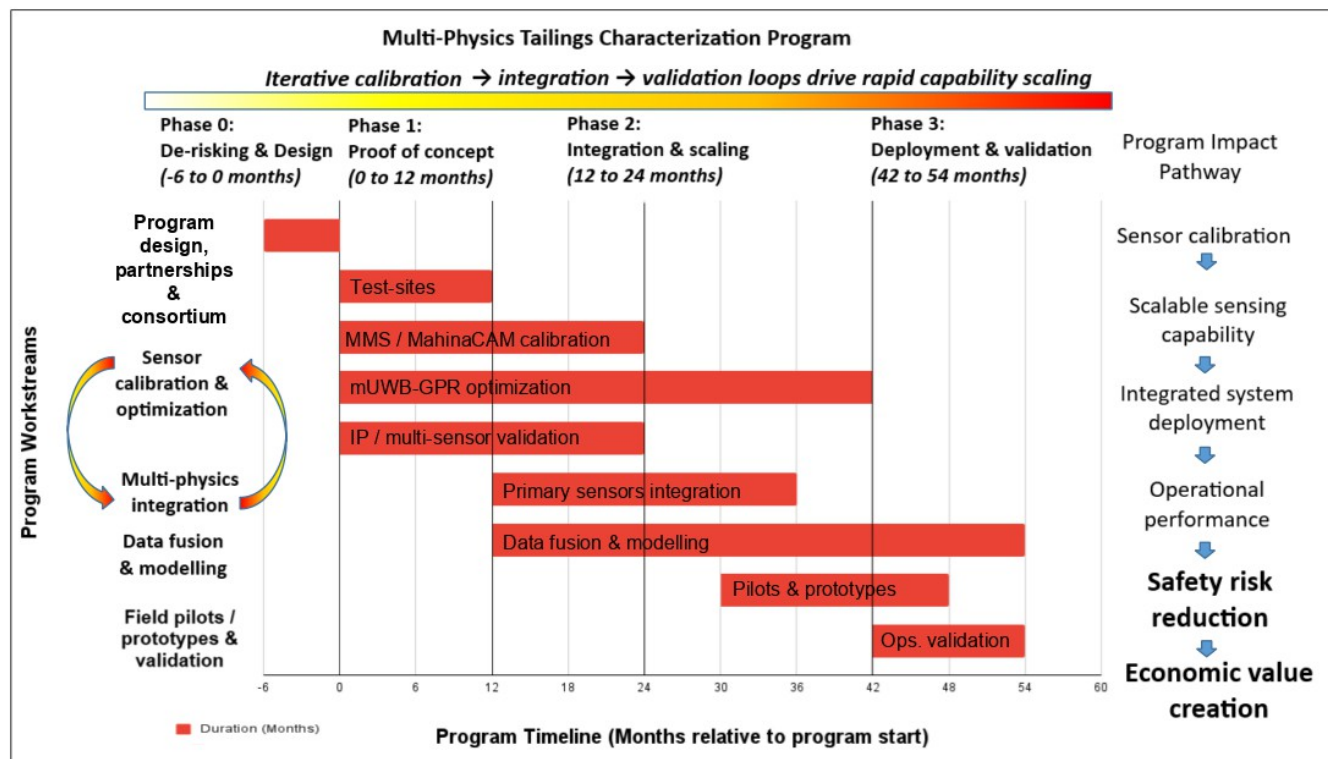


Figure A3.1 Program roadmap illustrating phased development and deployment of the proposed multi-physics tailings characterization architecture, progressing from calibration and integration to field validation, operational deployment, and safety and economic outcomes.

- Appendix 4 – Stakeholder Landscape and Partnerships

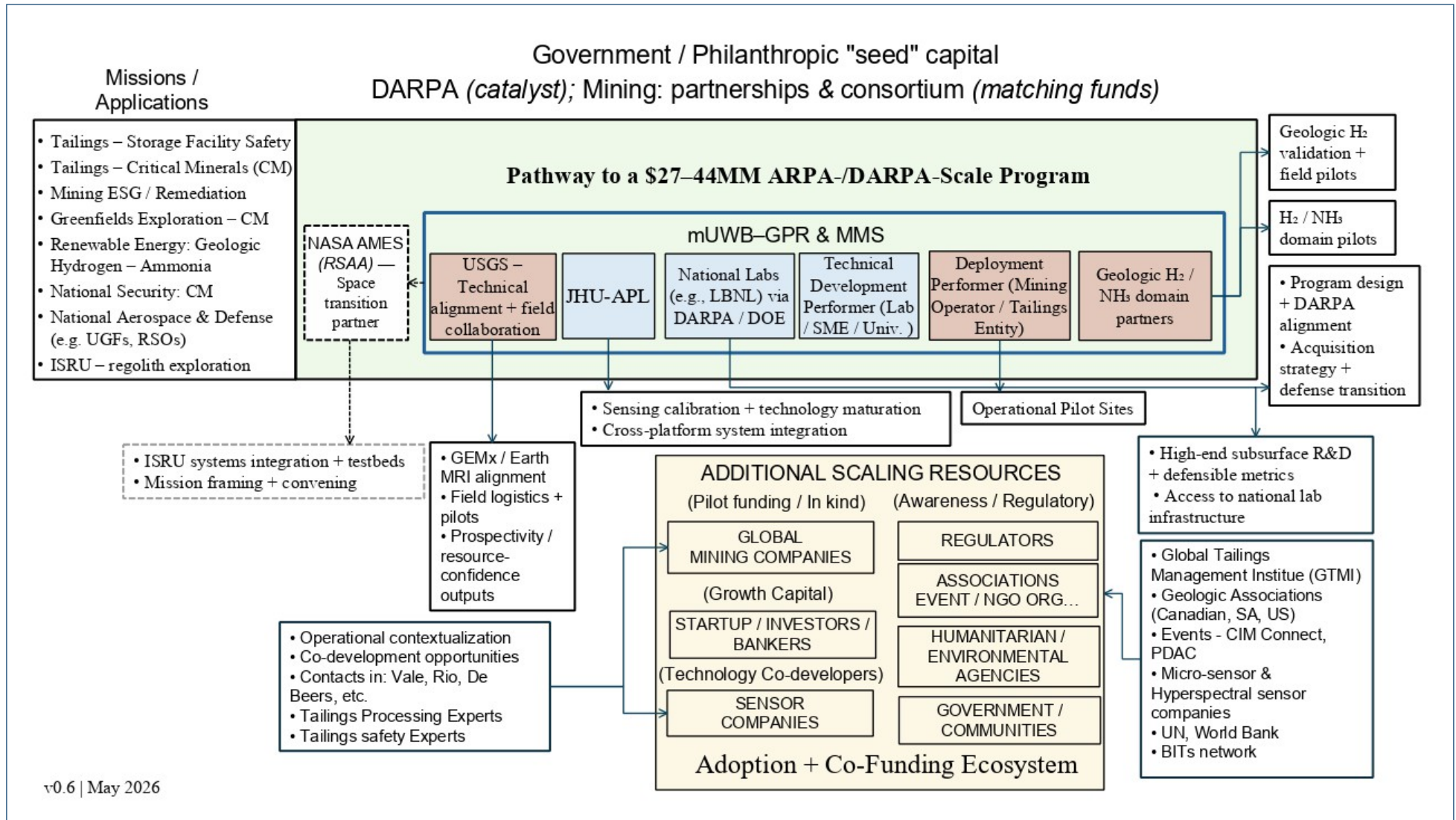


Figure A4.1 Stakeholder landscape for the proposed volumetric tailings characterization program, illustrating relationships between funding sources, coordinating organizations, research performers, industry partners, and data collaborators.

- Appendix 5 – Technical Notes and Sensing Modalities

Table A5.1 Factors Affecting Sensor Selection in Conductive, Iron-Rich Tailings

Hypothesized Sensor Applicability (subject to experimental validation)			
Key challenge: Deep (~90 m), saturated, iron-rich tailings create a sensing “blind zone” for most existing technologies.			
Technique	Why	Comment (Depth / Fe–H ₂ O constraints)	Primary Measurement
Microseismic	Requires cracking/friction; Static liquefaction produces no measurable dynamic signal	Not limited by depth or Fe–H ₂ O, but failure mode produces no signal	Dynamic energy release (acoustic emissions, fracturing)
DAS / Seismic Interferometry	Measures strain/vibrations; No dynamic signal pre-failure	Depth OK; not Fe-sensitive; failure mode produces no measurable strain signal prior to collapse	Strain / vibration along fiber (wavefield changes)
MASW / ReMI / HVSR	Bulk stiffness only; not continuous; Insensitive to localized creep	Poor resolution at depth (~90 m); indirect sensitivity to saturation	Shear-wave velocity (Vs) → bulk stiffness
ANI	Detects stiffness change; Requires repeat surveys; subtle signal	Resolution degrades at depth (~90 m); may capture trends, not localized instability	Changes in seismic velocity / stiffness over time
InSAR	Supplementary (surface-only)	Not depth-limited; blind to subsurface processes	Surface displacement (mm-scale deformation)
Thermal / Multispectral / Drone	Surface-only; Cannot detect internal saturation or creep	Not depth-limited; only useful for visible seepage	Surface temperature / reflectance (moisture, alteration)
Microgravity (MG)	Detects large density changes only	Very poor resolution at depth; insensitive to early liquefaction	Density variations (mass distribution)
SRT (Seismic Refraction)	Identifies large structural contrasts	Coarse resolution at depth; not sensitive to evolving failure state	Seismic velocity contrasts (layering / structure)
SBP	Only applicable in ponded conditions	Context-specific; not applicable to dry/wet TSF instability like Brumadinho	Acoustic impedance (sediment layering in water)
Magnetic Resonance	Direct water detection	Depth limited; affected by magnetic minerals (Fe); expensive / slow	Direct groundwater / water content (hydrogen response)
Borehole Radar (GPR)	Local structural detection	Strongly degraded by water + Fe conductivity; limited range; not scalable	Dielectric contrast (structure, voids, moisture)
VSP / Cross-hole Seismic	High-resolution imaging near boreholes	Good at depth; not Fe-sensitive; not scalable for full facility monitoring	High-resolution seismic velocity between boreholes
Muon Tomography (Muon Vision)	Passive density imaging; Not affected by Fe conductivity; May help with saturation/density-domain tracking	Vendor-reported pad-scale depth of ~50–100 m with high sensitivity (vertical resolution < 1 m; lateral resolution: ~2–5 m), but practical resolution at ~100 m is geometry- and time-dependent; better treated as promising, not assumed sub-metre at depth. Better treated as a calibration / validation tool rather than assumed high-resolution monitoring at depth	Density contrast (muon attenuation)
ERT / ERI / 3D ERT	Detects saturation/seepage (key driver)	Resolution degrades strongly at ~90 m; Sensitive to Fe–H ₂ O conductivity; interpretation becomes non-unique at depth	Electrical conductivity / resistivity (saturation, seepage)
Ground EM	Conductivity responds to moisture pathways	Coarse resolution at depth; strongly influenced by conductive Fe–H ₂ O system	Electromagnetic conductivity (moisture pathways)
Applicability reflects expected performance under deep (~90 m), saturated, iron-rich tailings conditions and will be validated through integrated multi-sensor field calibration.			
This table defines a working hypothesis for sensor applicability, forming the basis for Phase 1 calibration and subsequent development of scalable, proxy-based monitoring architectures.			
✗ Not suitable for early warning ● Supplementary (low resolution) ⊘ Not applicable ● Calibration / validation ⬆ Potential monitoring tool ✔			

Note: While induced polarization (IP) is a primary calibration modality, the program evaluates complementary methods as needed. This is critical in iron-rich, saturated tailings where conductivity effects can limit individual techniques; a multi-modal approach ensures robust characterization.

Acronyms

Modular Micro-Sensor (MMS) Spectral System | MhCAM — MahinaCAM | mUWB-GPR — Multistatic Ultra-Wideband Ground-Penetrating Radar | IP — Induced Polarization | ERT / EM — Electrical and electromagnetic methods (fluids, saturation, conductivity) | Seismic (MASW / ReMI / HVSR / refraction) — Structure, stiffness | DAS — Distributed Acoustic Sensing (fiber-optic seismic monitoring) | InSAR — Interferometric Synthetic Aperture Radar (deformation, stability) | Borehole methods (GPR / logging / XRF / LIBS) — Ground truth and calibration | Microseismic — Dynamic behaviour, failure signals | MG — Microgravity (density variations)

- Appendix 6 – Letters of Intent
A6.1 – HZDR

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02/05/2026

Letter of Intent

To whom it may concern,

I am pleased to express my interest in collaborating on the project "Volumetric Tailings Characterization: Safety & Valorization" as part of the Renaissance Philanthropy Program.

My research focuses on satellite- and UAV-based hyperspectral remote sensing for mine waste characterization. Using imaging spectroscopy, I develop methods to map tailings surface mineralogy, build multitemporal time series of compositional change, and integrate surface observations with subsurface geophysical data to derive three-dimensional material models. These capabilities are directly relevant to the project's core themes of valorization (mineral resource assessment and reprocessing planning) and safety (stability-relevant change detection).

Within the consortium, satellite hyperspectral data would provide the spatially continuous surface characterization layer that links the various ground-based and subsurface sensing methods and feeds into the data integration and valorization workstreams.

I look forward to discussing this further.

Sincerely,



Moritz Kirsch

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A6.2 – JHU-APL



May 6, 2026

Letter of Intent

To Whom It May Concern:

I am pleased to express my interest in and intent to collaborate on the project "Volumetric Tailings Characterization: Safety & Valorization" as part of the Renaissance Philanthropy Program.

My research focuses on spacecraft and airborne hyperspectral and multispectral imaging for surface composition characterization, related instrument development, and laboratory verification. I develop multispectral and hyperspectral techniques to address critical questions related to surface composition to infer origin, evolution, and processes responsible for the observed phenomena and to identify specific targets of interest in a complex and varied environment. I use laboratory experimentation to validate remote sensing approaches and develop remote sensing instrumentation for conducting tailored measurements. These capabilities are directly relevant to the project's goals of transforming tailings from unknown risks into mapped, measurable, and valuable resources.

Within the consortium, airborne and satellite multispectral and hyperspectral data would provide spatially continuous surface characterization layer that links the various ground-based and subsurface sensing methods and feeds into the data integration and valorization workstreams.

I look forward to participating!

Sincerely,

A handwritten signature in black ink, appearing to read 'Charles A. Hibbitts'.

Charles A. Hibbitts, PhD
Principal Professional Staff
Space Exploration Sector, JHUAPL

- Appendix 6 – Letters of Intent
A6.3 IMDEX

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IMDEX™

Dr Iuma Martinez

Team Leader, BiTS

Borehole & In-Situ Technology Strategy Program

Re: Letter of Intent – IMDEX Support for the BiTS Program

Dear Dr Martinez,

I am pleased to write in support of the Borehole & In-Situ Technology Strategy (BiTS) program and to express IMDEX's intent to contribute to its objectives. IMDEX is a leading provider of downhole sensing and geoscience data solutions to the global minerals industry, and we believe our in-situ geochemical analysis technology and advisory capability are directly aligned with the sensing and characterisation goals of the BiTS program.

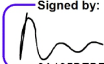
IMDEX is developing IBERIA — an in-situ elemental characterisation sensor operating at TRL 5. IBERIA is a downhole, wireline-deployed XRF sensor that delivers continuous, real-time elemental logs without radioactive sources or laboratory turnaround. Its non-radioactive architecture, continuous log output, and software-defined analytics layer make it a strong candidate for integration within the multi-sensor borehole characterisation systems envisioned under BiTS. We see clear potential for IBERIA to contribute to BiTS workstreams including in-situ solution chemistry monitoring, elemental profiling in poor core-recovery environments such as tailings, and as a calibration and proxy sensing node within the broader multi-physics sensing stack.

Beyond our sensing technology, IMDEX is prepared to contribute technical expertise and strategic advisory support to the program. Our team brings deep domain knowledge in downhole instrumentation, geoscience data workflows, and the practical deployment constraints of field-ready sensing systems — experience that we believe will add value to the program's development roadmap and its ambition to reduce characterisation cost and time by an order of magnitude.

IMDEX is supportive of the BiTS program's mission to move tailings characterisation from sparse, surface-based sampling toward continuous, volumetric 3D understanding. We recognise the significance of this challenge — both for reducing the risk of catastrophic tailings failures and for unlocking the critical mineral value held in historical tailings deposits globally.

We look forward to discussing how IMDEX can best contribute to the consortium and to the program's next phases. Please do not hesitate to contact us to progress this further.

Yours sincerely,

Signed by:

8A1358EBEA14440...

Dave Lawie

Chief Geoscientist

IMDEX Limited

- Appendix 6 – Letters of Intent
A6.4 – Telops / Exosens



Quebec, 5 May 2026

Object : Letter of Intent for Volumetric Tailings Characterization: Safety & Valorization Project

To whom it may concern,

I am writing to express our intent to participate to the *Volumetric Tailings Characterization: Safety & Valorization* project submitted to the *Renaissance Philanthropy Program*.

Telops/Exosens is a recognized industry leader in ground-based and airborne high-performance thermal infrared hyperspectral sensing technologies. We are pleased that our thermal infrared hyperspectral cameras can bring unique capabilities that will serve the core objectives of this project. The identification of key geological minerals based on their thermal infrared spectral emissivity fingerprints will serve the tailings valorization objective. Furthermore, the real-time detection and identification of gaseous chemical leaks will be critical to ensure health and safety of the surrounding environment.

We look forward to working with the consortium on this project.

Sincerely,

A handwritten signature in blue ink that reads "Vincent Farley".

Vincent Farley
Business Development Manager
v.farley@exosens.com
+1 (418) 264-7805

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A6.5 Endolith



7 May 2026

To whom it may concern,

Endolith is pleased to provide this non-binding letter of support for luma Martinez Germano's BiTS concept note. Endolith develops and applies bio-enabled mineral processing and critical mineral recovery technologies, with practical experience in mineralogical assessment, leach testing, and translation of laboratory results into commercially relevant process metrics.

Subject to project scope, timing, and commercial agreement, Endolith would be interested in supporting the proposed work by advising on and/or conducting mineralogical and leach test work relevant to BiTS. This support could include reviewing feed material characteristics, designing fit-for-purpose leach tests, generating recovery and acid consumption data, and helping interpret results in the context of process performance.

Endolith will also assist in translating leach test outcomes into a preliminary techno-economic assessment and business case, including indicative recovery assumptions, acid consumption estimates, and key operating considerations. Our goal would be to help the project team connect technical test results to practical recovery pathways and decision-ready commercial metrics.

We believe the proposed work is well aligned with Endolith's technical capabilities and interest in advancing more efficient approaches for critical mineral recovery. We would welcome the opportunity to discuss the project in more detail as the concept develops. This letter is provided for concept note support only and does not create any binding commitment by Endolith.

Sincerely,

A Arthur

A Arthur (May 8, 2026 10:37:19 GMT+10)

Andrew Arthur

CFO/CCO – Endolith

aarthur@endolithmining.com

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A6.6 Hitotsubashi University



HITOTSUBASHI
UNIVERSITY

May 8, 2026

Letter of Intent

To whom it may concern,

I am pleased to express my interest in collaborating on the project "Volumetric Tailings Characterization: Safety & Valorization" as part of the Renaissance Philanthropy Program.

My expertise lies in resource and energy economics, with a focus on market analysis and policy science. In the context of mine waste characterization, my work involves estimating demand and supply of tailings reprocessing and providing strategic advice on the regulatory and policy frameworks necessary for sustainable mineral resource management.

Within the consortium, my role would be to provide the socio-economic and policy integration layer. By analyzing the market drivers and policy constraints, I can ensure that the technical outputs of the project are effectively utilized to inform investment decisions and shape robust policies.

I look forward to discussing how my economic and policy-oriented perspective can contribute to the success of this multidisciplinary initiative.

Sincerely,

Hide-Fumi Yokoo
Associate Professor, Graduate School of Economics, Hitotsubashi University
2-1 Naka, Kunitachi, Tokyo 186-8601, Japan
E-mail: hidefumi.yokoo@r.hit-u.ac.jp

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- A6.7 The University of British Columbia



THE UNIVERSITY OF BRITISH COLUMBIA

Norman B. Keevil Institute of Mining Engineering
Faculty of Applied Science

May 11, 2026

To whom it may concern,

RE: Letter of intent to support the program on 'Reducing tailings risk & unlocking value through scalable sensing'.

I am pleased to confirm my intent to support the research program led by Ms. Uma Martinez Germano on 'Reducing tailings risk and unlocking value through scalable sensing' which forms part of the Renaissance Philanthropy Program.

I am an assistant professor at the Norman B. Keevil Institute of Mining at the University of British Columbia and holder of the Teck Professorship in Tailings Management and Innovation. I have been researching the geotechnical risks of tailings storage facilities for over 17 years. I have experience in laboratory and in situ geotechnical testing of tailings, use of remote sensing to assess geotechnical risk and was one of two geotechnical engineers that investigated the geotechnical causes of failure of the Jagersfontein TSF in South Africa. I look forward to supporting Ms. Germano's research program with my experience on the geotechnical risks of tailings.

Should you have any questions about my background or my planned involvement in the research program, please feel free to contact me using the email address listed below my signature.

Sincerely,

Prof. **Luis A. Torres-Cruz** CEng, MICE

Teck Professor in Tailings Management and Innovation

Faculty of Applied Science | Mining Engineering

The University of British Columbia | Vancouver Campus | Musqueam Traditional Territory

507 – 6350 Stores Road | Vancouver BC | V6T 1Z4 Canada

Luis.TorresCruz@ubc.ca

<https://mining.ubc.ca/person/luis-alberto-torres-cruz/>

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A6.8 USC Viterbi School of Engineering

Professor Mahta Moghaddam
Ming Hsieh Department of Electrical Engineering
Director, Microwave Systems, Sensors, and Imaging Lab
MiXIL

11 May 2026

Letter of Intent

To whom it may concern,

I am delighted to express my interest in collaborating on the project "Volumetric Tailings Characterization: Safety & Valorization" as part of the Renaissance Philanthropy Program.

I have over three decades of experience in microwave sensing for water-resource characterization, ground-based wireless networks, and autonomous UAV-based and distributed sensor networks for subsurface exploration. Prior to my switch to academia in 2003, and while at the NASA Jet Propulsion Laboratory (JPL), I was also Systems Engineer for the Cassini Radar and served as Science Chair of the JPL Team X (Advanced Mission Studies Team). Most recently, my group's research has focused on the development of new radar instrument and measurement technologies, including ultrawideband software-defined radar, for subsurface and subcanopy characterization, root-zone soil moisture, ground water, and permafrost applications, and geophysical retrievals using signal-of-opportunity reflectometry.

Within the proposed Consortium, I look forward to providing expertise for the development of microwave sensors at long wavelengths for deep, yet high-resolution, probing beneath the surface, to provide three-dimensional maps of material characteristics. The tension between sensing wavelength and resolution has been a long-standing challenge in non-contact mapping techniques. By taking advantage of multi-node UAV-based tomographic techniques, we believe that this can be overcome to a large extent. In collaboration with the Consortium's team of expert, we will define a tractable and systematic approach for simultaneously solving the resolution and specificity challenges.

Please do not hesitate to contact me with any questions.

Best regards,

A handwritten signature in black ink, appearing to read "Mahta Moghaddam".

Mahta Moghaddam
Distinguished Professor
Ming Hsieh Endowed Chair in Electrical and Computer Engineering
PI, Microwave Systems, Sensors, and Imaging Lab (MiXIL)
USC Viterbi School of Engineering
mahta@usc.edu



- Appendix 6 – Letters of Intent
A6.9 Channel Systems



Channel Systems

204-753-5190



info@channelsystems.ca



S2-24 Aberdeen Avenue
Pinawa, MB Canada R0E1L0



May 11, 2026

RE: Letter of Intent – Collaboration Interest for the “Volumetric Tailings Characterization: Safety & Valorization” Project under the Renaissance Philanthropy Program

To Whom it May Concern,

Channel Systems is pleased to express its interest in participating as a collaborative industry partner in the proposed “Volumetric Tailings Characterization: Safety & Valorization” initiative being developed under the Renaissance philanthropy program.

We understand the project aims to advance the integration of multiple sensing and analytical modalities to develop a coherent three-dimensional understanding of tailings systems, supporting both safety assessment and resource valorization objectives. Channel Systems recognizes the strong potential of this multidisciplinary approach and is interested in contributing within the broader multi-sensor, multi-partner framework being envisioned.

Channel Systems specializes in advanced imaging and sensing technologies, with expertise spanning hyperspectral, infrared, multispectral, and visible-spectrum imaging systems. Within the context of this project, we foresee potential contributions in areas including:

- Advisory input on hyperspectral and complementary sensor system selection and configuration
- Guidance on integration of spectral sensing technologies within the broader sensing and data acquisition framework
- Support for targeted laboratory sample scanning and/or field-based spectral data acquisition activities where appropriate
- Input into calibration, processing, and interpretation workflows related to spectral response and material property analysis
- Technical consultation regarding multi-modal sensing strategies integrating hyperspectral, thermal, multispectral, and/or visible imaging approaches
- Collaboration on approaches for integrating spectral datasets with other sensing modalities to support volumetric and spatial characterization objectives

We believe the proposed initiative represents an innovative and important opportunity to combine complementary technologies and expertise across industry and research partners. Channel Systems would welcome the opportunity to continue discussions regarding potential collaboration and participation as the project framework develops further.

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A6.9 Channel Systems (continued)



Channel Systems

204-753-5190



info@channelsystems.ca



S2-24 Aberdeen Avenue
Pinawa, MB Canada R0E1L0



This Letter of Intent is non-binding and does not constitute a formal commitment. Any future participation would be subject to further technical discussions and mutually agreed project terms.

We appreciate the opportunity to be considered as a potential collaborator and look forward to continued discussions about how we can best contribute to the consortium on this project!

Kind Regards,

Kate Kaminski
Director of Marketing & Administration
S2-24 Aberdeen Avenue
Pinawa, MB Canada R0E 1L0
T: 1-204-753-5190
E: Kate.kaminski@channelsystems.ca

ⁱ In this context, “safety” refers primarily to geotechnical and environmental risk associated with instability, saturation, liquefaction potential, and structural change within tailings systems.

ⁱⁱ Adaptive calibration refers to the selective use of complementary methods (e.g., IP, ERT/EM, seismic) as needed for a given site to constrain and refine interpretation of multi-sensor data.

ⁱⁱⁱ Induced polarization (IP) is a geophysical method that measures the ability of subsurface materials to store and release electrical charge, providing sensitivity to mineralogy (e.g., sulfides, clays) and supporting interpretation of material composition and distribution.

^{iv} In mining, consortium-based models are often not effective at accelerating commercial deployment. Constraints around data sharing, intellectual property, operational risk, and site access can slow decision-making and adoption. As a result, direct, site-specific partnerships with operators are more effective for calibration, validation, and real-world deployment of integrated sensing systems.

^v Through Helmholtz-Zentrum Dresden-Rossendorf (HZDR), we are connected to a global mining R&D network including Nordic Strategy Partners AB | GFZ | CTTC | Babeş-Bolyai University | TheiaX GmbH | GeoKinesia | Boliden Mineral AB | Advanced Mining Technology Center | Codelco | Cupru Min S.A. | FQM Trident Ltd.